

BSCHE Graduate Written Examination – Fall 2025

Written examination conducted with graduating seniors in Chemical Engineering

Name: Qussay Suleiman

Question 1

Career plans. (Scoring rubric was designed to assess potential for life-long learning.)

Where do you see yourself in 5-10 years, and what steps will you take to get there?

I see myself being a reliable engineer at my company and possibly managing some smaller projects. I will make sure to continue refining my knowledge and expertise to not be a mediocre engineer who simply does what they're told.

How have you researched your career plans and options (companies, graduate schools, PE certification, etc.)?

I have applied for many jobs through handshake, LinkedIn, and company websites. I don't plan on pursuing graduate school until years after graduation, at a time when my position is stable within the company and I can slowly pursue post graduate education.

Does your prospective employer sponsor master's degree or MBA? Under which terms?

I do not have an offer yet, but I'm hoping they do sponsor a master's degree or MBA.

Question 2

Professional ethics and ethical ramifications of technology.

Can you name a recent news item or significant piece of legislation that impacts on technology or engineering, and comment on its ethical ramifications?

A recent example is the increasing government focus on regulating industrial emissions and tightening rules on PFAS (“forever chemicals”). This affects chemical engineers because industries must balance production needs with environmental protection. The ethical issue is ensuring companies do not ignore long-term environmental harm in favor of short-term economic gain. Engineers have a responsibility to design safer processes and support decisions that reduce pollution, even if they require extra cost or effort.

Have you experienced an ethical dilemma in your educational or work (e.g., co-op) experience?

One ethical dilemma I faced in school was during a group project when a team member suggested cutting corners on a simulation to save time. The results would have looked acceptable, but they wouldn’t have been honest or representative of the real system. I had to decide whether to go along with it or push for doing the work properly. I chose to speak up and fixed the simulation myself, because accurate engineering work matters for safety and integrity.

How would you approach an ethical dilemma? Where or to whom would you turn for guidance? Can ethical problems be approached in terms of a systematic methodology, or are you guided more by gut reaction, personal experience, upbringing, etc.?

If I encountered an ethical dilemma, I would first step back and look at the situation objectively. I would consider safety, honesty, and the potential impact on the public or environment. I would also consult reliable guidance, such as the AIChE Code of Ethics, a professor, a supervisor, or someone with more experience. I try to approach ethical problems using a combination of systematic thinking and personal values. My upbringing and past experiences matter, but I also believe engineers must follow clear standards and not rely only on gut reaction.

Are you familiar with the AIChE Code of Ethics? Can you name specific provisions in it?

Yes. One important provision is to “hold paramount the safety, health, and welfare of the public and protect the environment in performance of professional duties.” Another provision is to be honest and transparent in all professional communications. These matter because chemical engineers work on processes that directly affect people and the environment, so ethical decisions have real consequences.

Which courses at UIC have contributed most to your views on professional ethics? How?

Courses like Chemical engineering lab, Senior Design, and other lab courses at UIC influenced my view of professional ethics the most. These classes emphasize safety, environmental impact, accurate reporting of data, and responsibility in engineering decisions. Working on real-world design problems showed me how ethics is connected to engineering practice, not just theory.

Question 3

Working in multidisciplinary teams

Can you name two other types of engineers that you might find yourself working with?

I could work with biomedical engineers or electrical engineers.

In what context or application would you be collaborating with them?

I can work with biomedical engineers if I work in an R&D position or pharmaceutical company, and electrical engineers if I work in a plant, since they would be needed to install and inspect equipment as well.

Question 4

Numerical methods and computational proficiency (Excel, MATLAB or VBA): choose one problem.

- (a) Generate a graph of the solution of the following ODE initial value problem.

$$\frac{dx}{dt} = \frac{1}{\tau} \left[x_{\text{in}}(t) - x(t) \right] - kx(t)^2, \quad x(0) = 2$$
$$x_{\text{in}}(t) = 1 + \varepsilon \sin(\omega t)$$

with the constants $\tau = 3.4$, $k = 1.2$, $\varepsilon = 0.1$ and $\omega = 5$. Can you think of a chemical engineering problem (or, at least, a course) in which such an ODE would arise?

- (b) Find a numerical value for the following definite integral.

$$\int_1^7 \frac{dx}{1 + \cos^3 x + \ln x}$$

b) To evaluate the integral in Excel, I used the trapezoidal numerical integration method. I first selected a number of intervals n and calculated the step size $h = (7 - 1)/n$. Then I created a column of x -values from 1 to 7, increasing by h , and computed the function

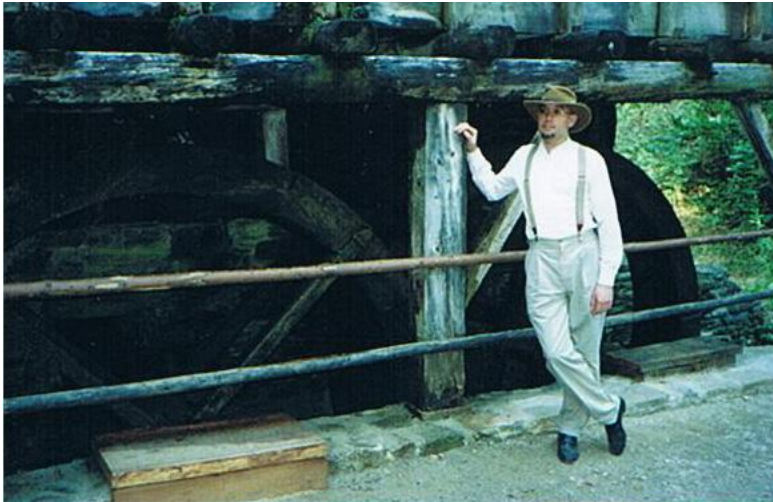
$$f(x) = \frac{1}{1 + \cos^3 x + \ln x}$$

for each x . I assigned a weight of 0.5 to the first and last points and 1 to all interior points (the trapezoidal rule). Finally, I multiplied each $f(x)$ by its weight, summed the results, and multiplied by h to obtain the numerical value of the integral. This produced an approximate answer of about **3.16**.

Question 5

Engineering Methodology Proficiency: Engineering estimate.

The waterwheel shown in the following three pictures was observed to make one complete revolution every 15 seconds. Estimate the power output (in units of horsepower)?





To estimate the power, I assume the wheel diameter is about 3 m (using the person for scale). Its circumference is therefore:

$$C = \pi D = 3.14 \times 3 = 9.4 \text{ m}$$

The wheel makes one revolution every 15 s, so the rim speed is:

$$v = C / 15 = 9.4 / 15 = 0.63 \text{ m/s}$$

A typical old mill waterwheel of this size is turned by a falling stream of water that delivers on the order of 1,000–2,000 N of torque-producing force on the buckets.

Use the midpoint $F = 1500 \text{ N}$.

$$\text{Mechanical power is: } P = F \times v = 1500 \times 0.63 \approx 945 \text{ W}$$

$$\text{Convert to horsepower: } 1 \text{ hp} = 746 \text{ W}$$

$$P = 945 / 746 = 1.3 \text{ hp}$$

Given uncertainty in diameter, water flow, and efficiency, the reasonable engineering estimate is: **1–2 horsepower**.